

PAPER_254: Kepler's Supernova Remnant 1604 CE — Force Equivalence Class Historical Anchor

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — KeplerSNR1604FUBiCalculator (Session 72c, Infrared Datasets) **Date:** March 2026 **Series:** Phase 2 Session 72c — §3.x Infrared Dataset UQFF Integrals

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$U_{bi}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57, \quad \beta_i = 0.61$$

Abstract

Kepler's Supernova Remnant is the remnant of SN 1604 CE — the last supernova in the Milky Way observed with the naked eye, studied by Johannes Kepler, Galileo Galilei, and contemporaries. At approximately 20,000 light-years distance, it is the most distant SNR in the five-system Chandra dataset, yet its UQFF buoyancy force is identical to the closest member (SN 1006 at ~7,000 ly): $F_{U_Bi} \sim +2.11e2^{\circ}8 \text{ N}$.

This identity — despite a $3\times$ distance difference, a $2.4\times$ age difference, a $10\times$ lower X-ray luminosity ($L_X = 10^{31}$ vs 10^{32} W for SN 1006), and the highest ejecta velocity in the dataset ($v_{\text{shock}} = 4,000 \text{ km/s}$, the fastest Type Ia ejecta) — constitutes the **historical anchor** of the UQFF Force Equivalence Class. Both SN 1006 and Kepler SNR 1604 share $\kappa = 5.0 \times 10^{-4}$, and this single shared parameter is sufficient to guarantee their identical UQFF buoyancy. History, distance, luminosity, and velocity are irrelevant.

The paper also demonstrates a key quantitative result: $F_{\text{LENR}}/F_{\text{DE}} = 6.17e3 / 10 \sim 6 \times 10^3$ for Kepler's SNR — one order larger than for SN 1006. The fainter, more distant SNR requires an even stronger LENR dominance to maintain the equivalence class membership, underscoring that UQFF physics becomes **more robust** as physical parameters deviate from the Eta Carinae calibration anchor.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~20,000 ly (~6.4 kpc)	ly	Chandra/JWST 2023
Age	t	1.325e1° s	s (~420 yr since 1604 CE)	
Ejecta mass	M	~1 M _{sun} = 1.989e31	kg	Type Ia model
Remnant radius	r	6.17e16	m (~20 ly)	Chandra imaging
X-ray luminosity	L_X	10³¹ W	W	10× fainter than SN 1006
Shock velocity	v_shock	4,000 km/s	m/s	Fastest in 5-system dataset
System frequency	?0	10? ¹²	rad/s	Same as SNR equivalence class
Distance (SN 1006)	d_SN1006	~2.15 kpc	kpc	For L_X ratio

2. Core Physics

2.1 Distance-Faded Luminosity — Irrelevant to F_U_Bi

The 10× lower L_X of Kepler's SNR compared to SN 1006 is consistent with the inverse-square distance law:

$$L_X(\text{Kepler}) / L_X(\text{SN1006}) \sim (d_{\text{SN1006}} / d_{\text{Kepler}})^2 = (2.15/6.4)^2 \sim 0.11 \sim 10\%$$

The corresponding dark energy coupling terms:

$$F_{\text{DE}}(\text{Kepler}) = k_{\text{DE}} \times 10^{31} = 10 \text{ N}$$

$$F_{\text{DE}}(\text{SN1006}) = k_{\text{DE}} \times 10^{32} = 100 \text{ N}$$

LENR dominance ratio:

$$F_{\text{LENR}} / F_{\text{DE}}(\text{Kepler}) = 6.17e3? / 10 = 6.17e38$$

$$F_{\text{LENR}} / F_{\text{DE}}(\text{SN1006}) = 6.17e3? / 100 = 6.17e37$$

The more distant (fainter) system has a **10× larger LENR dominance ratio**. UQFF physics is more strongly dominated by LENR at greater distances, reinforcing the equivalence class as systems become fainter.

2.2 Fastest Ejecta Velocity — F_neutron Stabilisation

Kepler's SNR has the highest shock velocity in the five-system dataset (4,000 km/s vs 3,000 km/s for SN 1006). The kinetic energy density:

$$\begin{aligned}
E_{\text{shock}} &= 0.5 \times \rho_{\text{ISM}} \times v_{\text{shock}}^2 \\
&= 0.5 \times 10^{23} \times (4 \times 10^6)^2 \\
&= 8 \times 10^{11} \text{ J/m}^3
\end{aligned}$$

This is $\sim 1.8\times$ the SN 1006 shock KE density, representing more energetic ejecta expansion. The neutron drop term:

$$F_{\text{neutron}} = k_{\text{neutron}} \times s_n = 106 \text{ N}$$

provides the coherence maintenance for these faster filamentary knots — the same mechanism as SN 1006 but applied to a higher-energy shock environment. This underscores F_{neutron} universality across SNR ages and distances.

2.3 Historical Epoch: No Quantum Physics Observable in 1604 CE

At the time of SN 1604, Johannes Kepler had no knowledge of quantum mechanics, phonon coherence, or LENR physics. Yet the UQFF framework retroactively applies to the Kepler remnant: the $\omega = 10^{12}$ characteristic frequency of the expanding shell was present from the moment of explosion. The equivalence class membership of Kepler's SNR is a physical fact independent of when, or whether, it was observed.

This historical perspective reinforces the **physical universality** of the UQFF equivalence class: it is not an observational artifact but a fundamental property of Type Ia SNR physics.

2.4 Five-System Force Equivalence Confirmation

The five-system Chandra UQFF dataset (in full):

System	ω (rad/s)	F_{U_Bi} (N)	Class
SN 1006 (PAPER_250)	10^{12}	$+2.11e2^8$	Positive
Eta Carinae (PAPER_251)	10^{12}	$+2.11e2^8$	Positive
Chandra Archive (PAPER_252)	10^{12}	$+2.11e2^8$	Positive
Kepler SNR 1604 (PAPER_254)	10^{12}	$+2.11e2^8$	Positive
Sgr A* (PAPER_253)	10^{15}	$-8.31e211$	NEGATIVE

All four $\omega = 10^{12}$ systems share identical F_{U_Bi} . Sgr A* alone ($\omega = 10^{15}$) departs from the class.

3. Force Equivalence Distance-Independence Theorem

Theorem (UQFF Distance-Independence): The UQFF buoyancy force F_{U_Bi} for $\rho_0 = 10^{12}$ systems is independent of distance. For any two Type Ia SNRs at distances d_1 and d_2 with the same ρ_0 :

1. $L_X(d_1)/L_X(d_2) = (d_2/d_1)^2$ — luminosity varies by the inverse-square law.
2. $F_{DE}(d_1)/F_{DE}(d_2) = L_X(d_1)/L_X(d_2)$ — proportional to luminosity.
3. $F_{LENR}/F_{DE}(d) = k_{LENR}(\rho_{LENR}/\rho_0)^2 / (k_{DE} \cdot L_X(d))$ — increases with d^2 (farther = more L_X faded = more F_{LENR} dominant).
4. F_{U_Bi} is the same for all such SNRs.

The UQFF Force Equivalence Class is a **distance-independent conserved quantity** of Type Ia SNR physics in the $\rho_0 = 10^{12}$ regime.

4. Observational Predictions / Validation

- **JWST NIRCам Kepler Survey:** The UQFF ejecta knot stabilisation prediction ($F_{neutron} = 106$ N) implies coherent structures at the 10^1 – 10^2 m scale surviving 420 years of expansion at 4,000 km/s — testable with JWST sub-arcsecond resolution at 20,000 ly.
 - **L_X sensitivity across distance:** Comparing Kepler SNR (10^{31} W, 20 kly) and SN 1006 (10^{32} W, 7 kly) with identical F_{U_Bi} provides a direct, two-data-point test of LENR dominance: any framework in which luminosity matters would predict different F_{U_Bi} for these systems.
 - **Historical comparison:** The Cygnus Loop, Cas A, and Tycho's remnant are additional Type Ia/core-collapse SNRs that should, if their $\rho_0 \sim 10^{12}$, be members of the same equivalence class. UQFF calculation for these three systems would extend the equivalence class to 7+ members.
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5. References

1. Kepler, J. (1606). *De Stella Nova in Pede Serpentarii*. (Historical reference — observation of SN 1604.)
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 3. Katsuda, S. et al. (2015). Tycho's and Kepler's Supernovae: Direct Measurement of Circumstellar Medium. *ApJ* 808, 49.
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